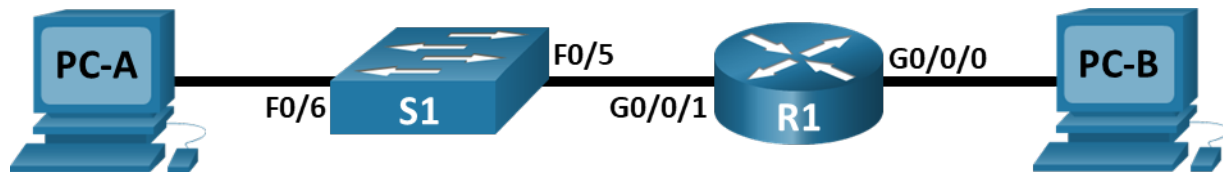


## Lab - Build a Switch and Router Network

### Topology



### Addressing Table

| Device | Interface | IP Address / Prefix   | Default Gateway |
|--------|-----------|-----------------------|-----------------|
| R1     | G0/0/0    | 192.168.1.1 /24       | N/A             |
|        |           | 2001:db8:acad:1::1/64 |                 |
|        |           | fe80::1               |                 |
|        | G0/0/1    | 192.168.0.1 /24       | N/A             |
|        |           | 2001:db8:acad::1/64   |                 |
|        |           | fe80::1               |                 |
| S1     | VLAN 1    | 192.168.0.2 /24       | 192.168.0.1     |
| PC-A   | NIC       | 192.168.0.3 /24       | 192.168.0.1     |
|        |           | 2001:db8:acad::3/64   | fe80::1         |
| PC-B   | NIC       | 192.168.1.3 /24       | 192.168.1.1     |
|        |           | 2001:db8:acad:1::3/64 | fe80::1         |

### Objectives

**Part 1: Set Up the Topology and Initialize Devices**

**Part 2: Configure Devices and Verify Connectivity**

### Background / Scenario

This is a comprehensive lab to review previously covered IOS commands. In this lab, you will cable the equipment as shown in the topology diagram. You will then configure the devices to match the addressing table.

After the devices have been configured and network connectivity has been verified, you will use IOS commands to retrieve information from the devices to answer questions about your network equipment.

This lab provides minimal assistance with the actual commands necessary to configure the router. Test your knowledge by trying to configure the devices without referring to the content or previous activities.

**Note:** The routers used with CCNA hands-on labs are Cisco 4221 with Cisco IOS XE Release 16.9.4 (universalk9 image). The switches used in the labs are Cisco Catalyst 2960s with Cisco IOS Release 15.2(2) (lanbasek9 image). Other routers, switches, and Cisco IOS versions can be used. Depending on the model

and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs. Refer to the Router Interface Summary Table at the end of the lab for the correct interface identifiers.

**Note:** Ensure that the routers and switches have been erased and have no startup configurations. Consult with your instructor for the procedure to initialize and reload a router and switch.

The **default bias** template used by the Switch Database Manager (SDM) does not provide IPv6 address capabilities. Verify that SDM is using either the **dual-ipv4-and-ipv6** template or the **lanbase-routing** template.

### Required Resources

- 1 Router (Cisco 4221 with Cisco IOS XE Release 16.9.4 universal image or comparable)
- 1 Switch (Cisco 2960 with Cisco IOS Release 15.2(2) lanbasek9 image or comparable)
- 2 PCs (Windows with a terminal emulation program, such as Tera Term)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet cables as shown in the topology

### Instructions

#### Part 1: Set Up Topology and Initialize Devices

##### Step 1: Cable the network as shown in the topology.

- Attach the devices shown in the topology diagram, and cable, as necessary.
- Power on all the devices in the topology.

#### Part 2: Configure Devices and Verify Connectivity

In Part 2, you will set up the network topology and configure basic settings, such as the interface IP addresses, device access, and passwords. Refer to the **Error! Reference source not found.** and **Error! Reference source not found.** at the beginning of this lab for device names and address information.

##### Step 1: Assign static IP information to the PC interfaces.

- Configure the IP address, subnet mask, and default gateway settings on PC-A.
- Configure the IP address, subnet mask, and default gateway settings on PC-B.

##### Step 2: Configure the router.

- Console into the router and enable privileged EXEC mode.
- Enter configuration mode.
- Assign a device name (R1) to the router.
- Disable DNS lookup to prevent the router from attempting to translate incorrectly entered commands as though they were host names.
- Assign **class** as the privileged EXEC encrypted password.
- Assign **cisco** as the console password and enable login.
- Assign **cisco** as the VTY password and enable login.
- Encrypt the plaintext passwords.
- Create a banner that warns anyone accessing the device that unauthorized access is prohibited.

- j. Configure and activate both interfaces on the router.
- k. Configure an interface description for each interface indicating which device is connected to it.
- l. To enable IPv6 routing, enter the command `ipv6 unicast-routing`.
- m. Set the clock on the router.

### Step 3: Configure the switch.

In this step, you will configure the hostname, the VLAN 1 interface and its default gateway.

- a. Console into the switch and enable privileged EXEC mode.
- b. Enter configuration mode.
- c. Assign a device name (S1) to the switch.
- d. Disable DNS lookup to prevent the router from attempting to translate incorrectly entered commands as though they were host names.
- e. Configure and activate the VLAN interface on the switch S1.
- f. Configure the default gateway for the switch S1.

### Step 4: Verify connectivity end-to-end connectivity.

- a. From PC-A, ping PC-B.
- b. From PC-B, ping PC-A.
- c. From PC-A, ping Switch.
- d. From PC-A, ping Router.
- e. From PC-B, ping Switch.
- f. From PC-B, ping Router.